EMBEDDED SYSTEM WITH IOT
EXPERIMENT NO. 5

IoT based Water Monitoring System

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Course/Section: CPE162P-4/E01

Group No.: INDIVIDUAL

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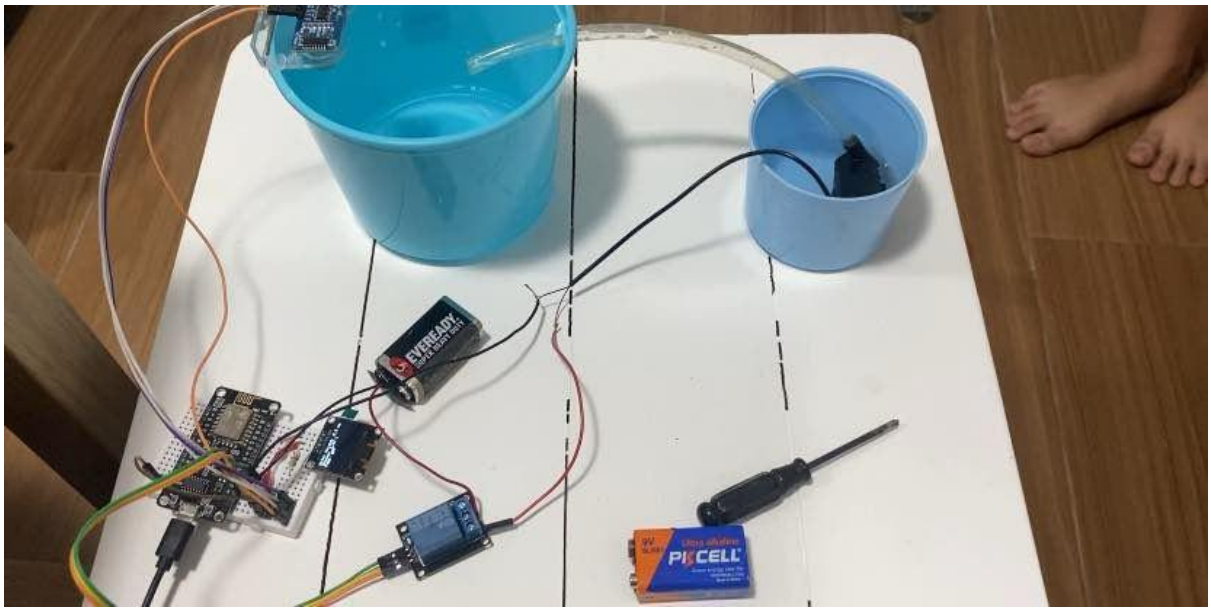
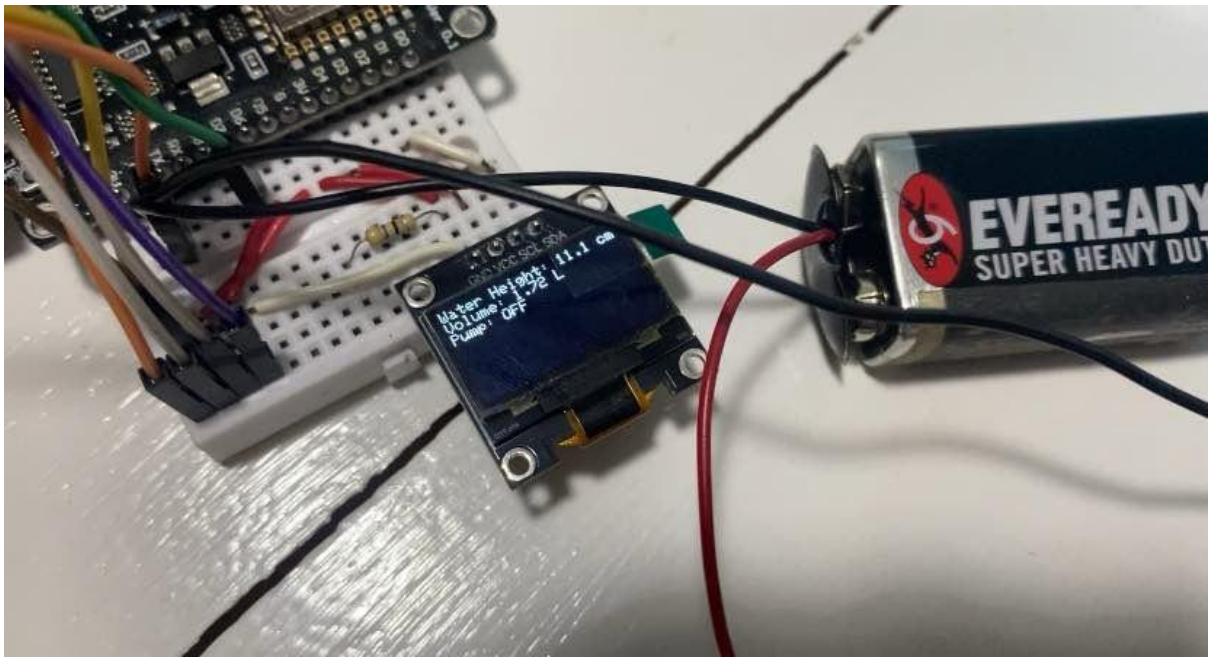
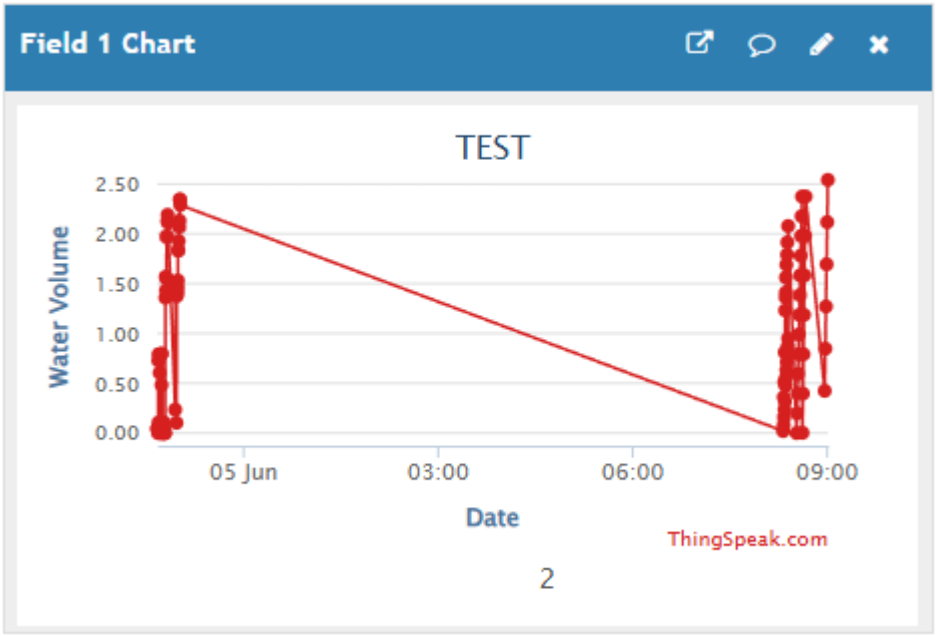
DISCUSSION


EVEREADY
PICELL


Water Height: 11.1 cm
Volume: 1.72 L
Pump: OFF
EVEREADY
SUPER HEAVY DUT

Experiment 5: IoT-based Water Monitoring System shows how to use an ESP8266 microcontroller, an OLED display, an ultrasonic sensor, and a relay-controlled water pump to create an automated water volume monitoring system. By measuring the distance between the sensor and the water's surface, the ultrasonic sensor determines the water height inside a cylindrical container. The dimensions of the cylinder are used to convert this distance to water volume. In order to simulate water flow, the relay cycles the pump ON for different testing seconds and OFF for twenty seconds. The OLED screen shows the pump status, volume in liters, and water height in real time. Every 15 seconds, water volume data is transmitted to ThingSpeak for remote monitoring and analysis.

Overall Data Entry (Analog Readings)


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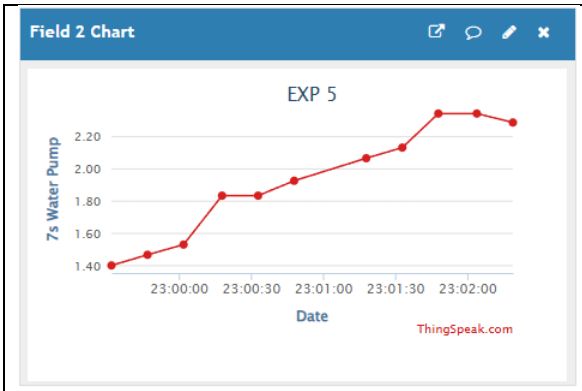
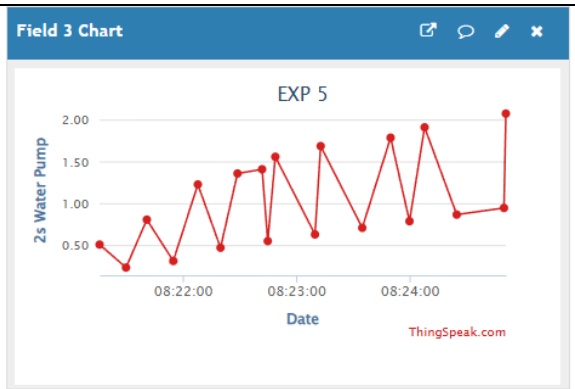
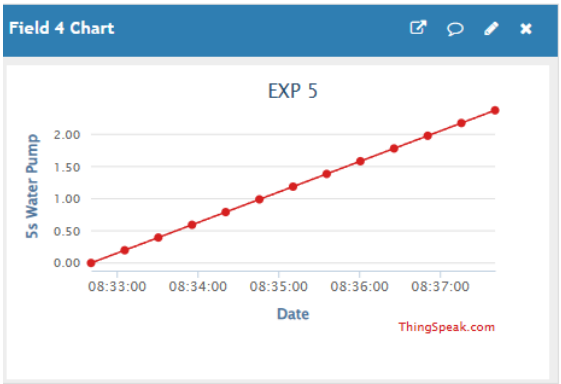
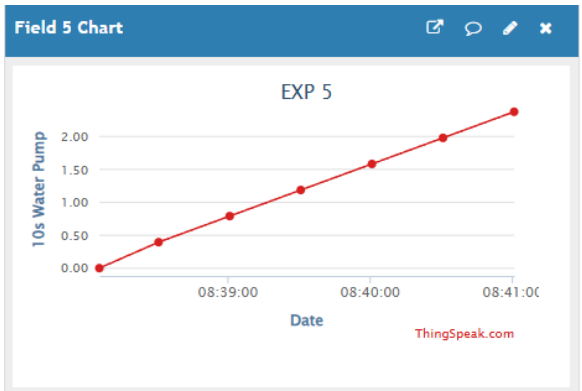
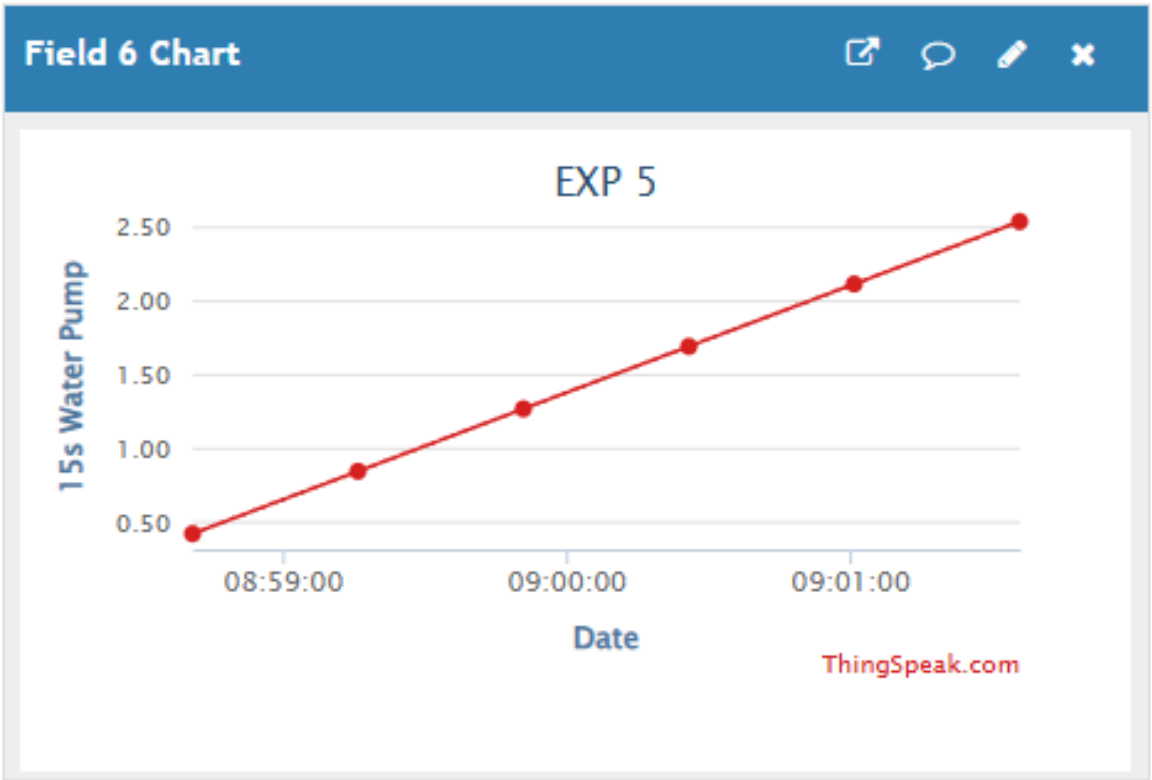
Field 1 illustrates the real-time water volume readings of an IoT-based water monitoring system using an ESP8266. Initially, water volume values fluctuate between 0.2 and 2.4 liters, reflecting the active operation of the relay-controlled water pump cycling ON and OFF. This consistent variation confirms accurate distance measurement by the ultrasonic sensor and proper conversion to volume.


Field 2 Chart
EXP 5
7s Water Pump
2.20
2.00
1.80
1.60
1.40
23:00:00
23:00:30
23:01:00
23:01:30
23:02:00
Date
ThingSpeak.com


Field 3 Chart
EXP 5
2s Water Pump
2.00
1.50
1.00
0.50
08:22:00
08:23:00
08:24:00
Date
ThingSpeak.com


Field 4 Chart
EXP 5
5s Water Pump
2.00
1.50
1.00
0.50
0.00
08:33:00
08:34:00
08:35:00
08:36:00
08:37:00
Date
ThingSpeak.com


Field 5 Chart
EXP 5
10s Water Pump
2.00
1.50
1.00
0.50
0.00
08:39:00
08:40:00
Date
ThingSpeak.com


Field 6 Chart
EXP 5
15s Water Pump
2.50
2.00
1.50
1.00
0.50
08:59:00
09:00:00
09:01:00
Date
ThingSpeak.com

Field Charts 2 through 6 from Experiment 5 illustrate the IoT-based water monitoring system s ability to accurately measure water volume across varying pump durations. Shorter durations (2s and 5s) show more frequent fluctuations or steady increases, while longer durations (10s and 15s) result in smooth, linear volume growth, confirming reliable pump control and sensor accuracy. The 7s pump chart shows gradual volume rise with slight variations, reflecting stable system performance. Overall, the charts demonstrate precise real-time tracking of water flow, consistent system responsiveness, and successful integration with ThingSpeak for effective remote monitoring.

CONCLUSION

Experiment 5 successfully achieved its primary objective of demonstrating how an IoT-based Water Monitoring System operates using embedded components such as the ESP8266, ultrasonic sensor, relay, and OLED display. Through precise distance measurements converted into real-time water volume readings, the system monitored the dynamic behavior of water flow in a cylindrical container. The relay-controlled pump, cycling on for different time durations, enabled the simulation of varying water flow rates, while the OLED provided live feedback on system status. The integration of these components showcased the fundamental

workings of IoT in a real-world water management application.

The experiment also reinforced key concepts from logic circuits and switching theory, as seen in the timed ON/OFF control of the relay using microcontroller logic. Practical applications of lessons from CPE160P and CPE161P were evident in the use of microcontrollers for decision-making, sensor interfacing, and real-time data handling. The experiment required configuring hardware logic for the relay pump, implementing sensor data acquisition, and structuring timing sequences for pump operation skills directly tied to foundational course learnings.

Furthermore, the successful use of ThingSpeak for 15-second interval data logging and

visualization fulfilled the objective of applying IoT platforms for data analysis.